

Homework 1 — Extended EDA and Data Pipeline for MILK10k

George Hanna · Computer Vision & Speech Recognition, EADA · October 2026 · code: package milk10k/; every figure and table is reproduced by python scripts/run_session2.py

MILK10k has 10,480 images (all 600×450 px) of 5,240 skin lesions, one dermoscopic and one clinical close-up per lesion. The target is diagnosis_1: 3,634 Malignant (69.4%), 1,483 Benign (28.3%) and 123 Indeterminate (2.3%) lesions; the 11-class diagnosis from training_gt.csv gives a finer view. **Main findings:** age and body site carry real but modest signal; six columns leak the label and must never be model inputs; simple colour statistics overlap heavily between classes (best single feature: ROC-AUC 0.66), so a model needs spatial patterns, not global colour.

1. Metadata vs. diagnosis (Part 1)

Method. Fields that are identical for both images of a lesion (all but image_type and image_manipulation) are analysed with one row per lesion (n = 5,240); using both images would count each lesion twice and overstate significance. Categorical fields: chi-square test of independence, with Cramér's V as effect size (0 = none, 1 = perfect). Age: Kruskal–Wallis test, a rank-based one-way ANOVA, because age comes in coarse 5-year bins, is skewed and spreads differently per class, which breaks ANOVA's normality and equal-variance assumptions (classic ANOVA agrees: $\eta^2 = 0.15$). With 5,240 lesions every p-value is below 0.001, so fields are ranked by effect size. Missing values form their own category, so informative missingness is measured too.

Column	Type	Missing	Unit	Test (3 classes)	V, 3 cl.	V, 11 cl.	Verdict
Candidate model inputs							
age_approx	numeric	0.4%	lesion	Kruskal–Wallis H = 586, p < 10 ^{−127}	0.31*	0.20*	moderate: strongest usable field
anatom_site_general	categorical (4)	37.3%	lesion	$\chi^2 = 260$, df 8	0.16	0.17	weak; missingness informative
image_manipulation	categorical (2)	0%	image	$\chi^2 = 202$, df 2	0.14	0.13	weak; acquisition proxy, bias risk
sex	categorical (2)	0%	lesion	$\chi^2 = 46$, df 2	0.09	0.12	negligible
anatom_site_special	categorical (2)	98.0%	lesion	$\chi^2 = 44$, df 4	0.06	0.14	negligible; nearly always missing
image_type	categorical (2)	0%	image	$\chi^2 = 0$, p = 1	0.00	0.00	none: one image of each per lesion
attribution, copyright_license	constant	0%	lesion	—	0	0	useless: a single value
Leak the label: never model inputs							
diagnosis_2 / _3 / _4	categorical	0 / 1.5 / 85.5%	lesion	$\chi^2 = 10,480 / 10,440 / 1,699$	1.00 / 1.00 / 0.40	0.95 / 0.91 / 0.37	finer levels of the label itself
melanocytic	boolean	77.2%	lesion	$\chi^2 = 897$, df 2	0.41	1.00	True exactly for NV and MEL
diagnosis_confirm_type, concomitant_biopsy	categorical / boolean	0%	lesion	$\chi^2 = 507$, df 2	0.31	0.31	identical fields: was it biopsied?

Table 1. Column audit and association with the label (identifiers isic_id, lesion_id and the target are excluded). V = Cramér's V. *For age the 17 age bins are treated as categories, to compare it on the same scale; Kruskal–Wallis effect size $\eta^2_H = 0.11$ (3 classes), 0.20 (11 classes).

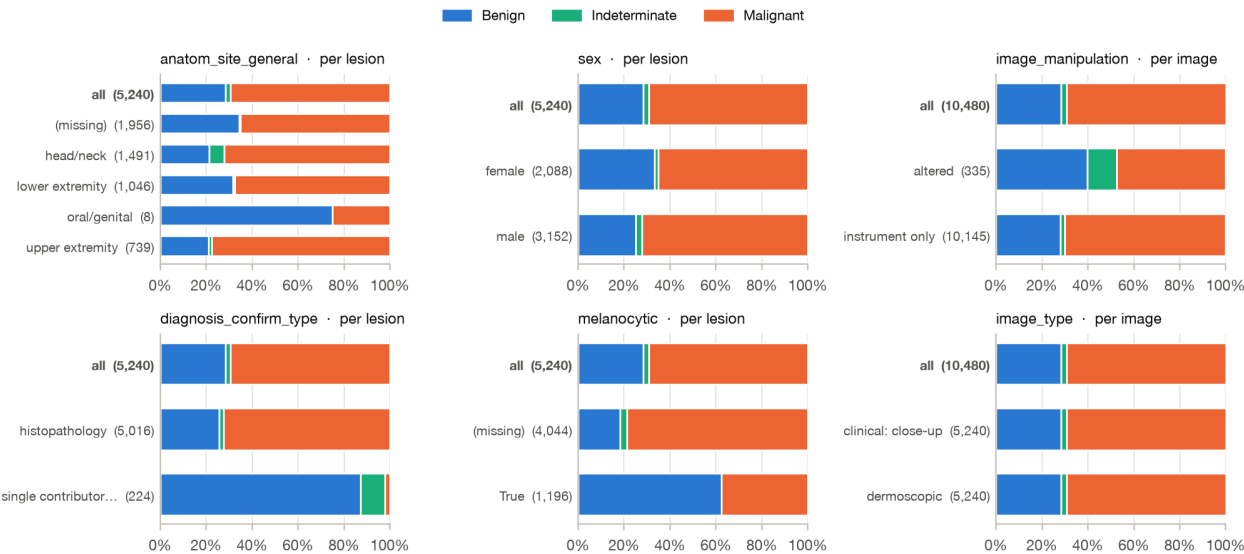


Figure 1. Label mix inside each category; the bold top bar is the overall mix. A field is informative when its bars differ from the top bar: clearly for the leaks (diagnosis_confirm_type, melanocytic) and image_manipulation, slightly for site and sex, not at all for image_type.

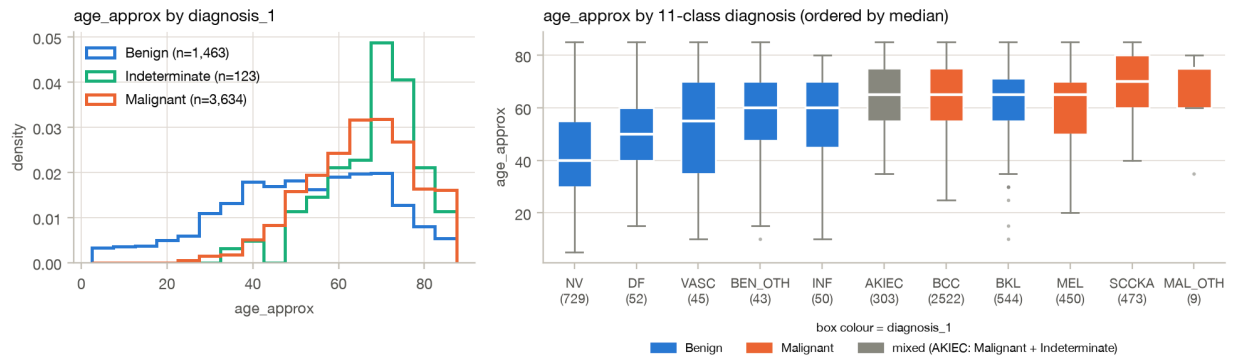


Figure 2. Benign lesions come from younger patients, driven by nevi (median 40 years) against 65–75 years for the malignant classes.

- **Most associated (usable): age** (median 40 years for NV, 65 for BCC and MEL, 70 for SCCKA) and **body site** (head/neck holds 6.6% Indeterminate lesions vs. $\leq 1.4\%$ elsewhere, consistent with actinic keratoses on sun-exposed skin; upper extremity is 77.7% Malignant); **image_manipulation** only with caution.
- **Least useful:** attribution, copyright_license (constant), image_type (uninformative by construction), sex (72% vs. 65% Malignant in men vs. women), anatom_site_special (98% missing).
- **Leakage:** diagnosis_2/3/4 are the label at a finer level; melanocytic is True exactly for nevi and melanomas; diagnosis_confirm_type and concomitant_biopsy record whether the lesion was biopsied, a decision taken *because* of the suspected diagnosis (87% of lesions confirmed by clinical assessment alone are Benign) and unknown when the model is used.
- **Bias and proxies:** “altered” images hold 12.8% Indeterminate lesions vs. 2.0% for “instrument only”; the field describes how an image was captured or edited and may stand for a clinic or protocol rather than the disease. A missing body site goes with more Benign lesions (34.5% vs. 28.3%), so missingness may encode the contributor. Age is legitimate, but a model could lean on it (“young patient → nevus”) instead of the image.

2. Colour and histograms across classes (Part 2)

Method. 100 lesions per diagnosis_1 class (seed 42, both images: 100 dermoscopic and 100 clinical images per class) and, for the finer view, 30 lesions per 11-class diagnosis (MAL_OTH: all 9). Per image: grey-level ($0.299R + 0.587G + 0.114B$) and R/G/B histograms normalised to sum to 1, so each image counts equally, plus the mean and standard deviation of each channel. Images are decoded at ≤ 256 px; normalised histograms barely change with resolution. Dermoscopic and clinical images are always analysed separately.

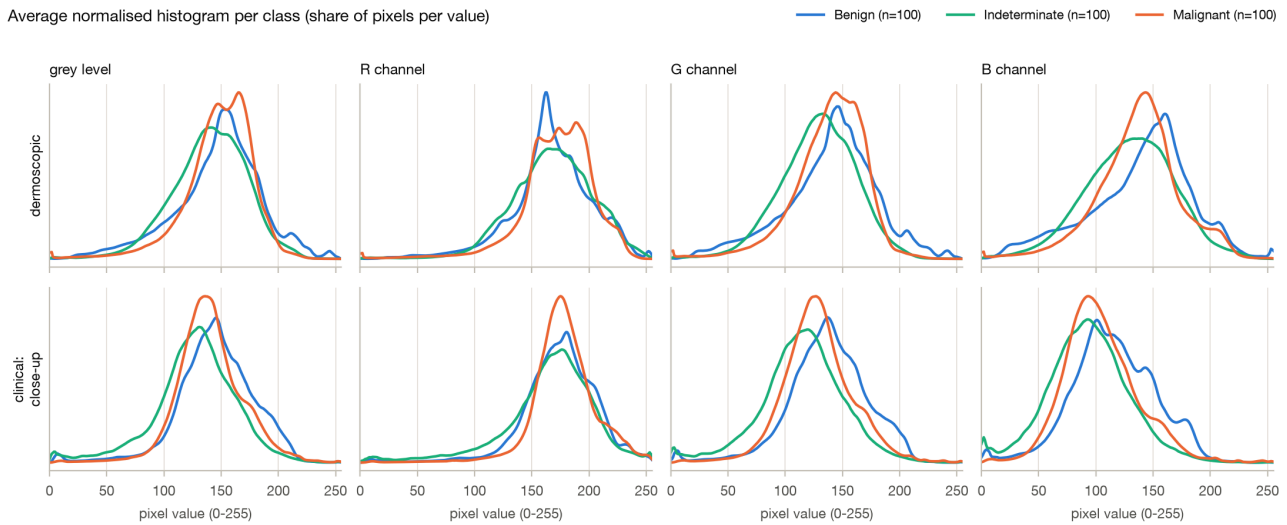


Figure 3. Average histograms per class. Within each image type the curves overlap heavily; the device effect (dermoscopic images are far bluer than clinical ones, right column) is larger than any class effect.

Image type	Class	n	Brightness	Mean R	Mean G	Mean B	Std R	Std G	Std B
dermoscopic	Benign	100	147.6 $\pm$ 20.2	167.0 $\pm$ 22.4	139.5 $\pm$ 21.2	138.9 $\pm$ 22.5	24.1	31.2	35.1
	Indeterminate	100	140.6 $\pm$ 21.8	167.8 $\pm$ 27.4	128.9 $\pm$ 20.5	129.8 $\pm$ 25.8	19.5	24.2	27.3
	Malignant	100	148.5 $\pm$ 16.3	172.2 $\pm$ 20.7	138.6 $\pm$ 16.7	137.1 $\pm$ 23.0	19.9	23.7	25.5
clinical close-up	Benign	100	145.7 $\pm$ 23.2	174.1 $\pm$ 22.8	137.1 $\pm$ 25.0	115.3 $\pm$ 26.7	20.1	22.2	22.3
	Indeterminate	100	127.0 $\pm$ 22.5	164.6 $\pm$ 26.2	114.4 $\pm$ 23.0	93.0 $\pm$ 22.7	26.6	25.9	25.5
	Malignant	100	140.7 $\pm$ 17.6	178.1 $\pm$ 18.5	128.6 $\pm$ 18.7	104.4 $\pm$ 21.9	18.1	22.2	22.4
ROC-AUC of each single feature, Malignant vs. Benign									
dermoscopic			0.52	0.58	0.50	0.46	0.37	0.35	0.34
clinical close-up			0.42	0.56	0.38	0.37	0.49	0.53	0.54

Table 2. Per-image colour statistics, 0–255 scale (mean $\pm$ standard deviation across images; the “Std” columns are means of the per-image standard deviations). ROC-AUC: 0.5 = complete overlap; below 0.5 = lower values in Malignant images.

Per-image colour statistics by 11-class diagnosis (box colour = diagnosis_1; grey = AKIEC, mixed)

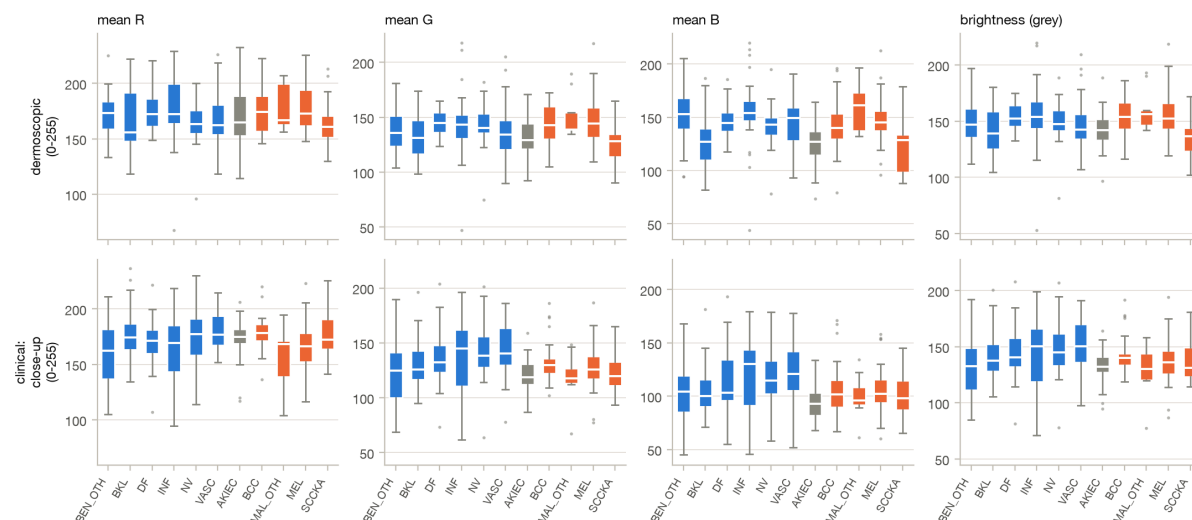


Figure 4. Per-image colour statistics by 11-class diagnosis (30 lesions per class; MAL_OTH 9). The boxes overlap for almost every pair of classes.

- **Within an image type, Benign and Malignant overlap almost completely:** mean brightness differs by ≤ 5 grey levels (dermoscopic 147.6 vs. 148.5, clinical 145.7 vs. 140.7) while images of one class spread by 16–23 levels. No single colour feature exceeds ROC-AUC 0.66 (dermoscopic Std B, 0.34 read the other way).
- **Indeterminate lesions** (all actinic keratoses) are the clearest outlier in clinical images: darker (127 vs. 141–146), with less green and blue, i.e. redder.
- **Malignant dermoscopic images vary less in colour** (Std B 25.5 vs. 35.1): benign classes include pigmented nevi, a dark lesion on light skin, while 69% of malignant lesions are BCC, typically pink and low-contrast. This is a class-composition effect, not a property of malignancy.
- **11 classes:** AKIEC, SCCA and BKL have the darkest, least blue dermoscopic images (median brightness 137–142, mean B 127–129, vs. 147–156 and 140–161 for most other classes).

Other explanations than “the diagnosis causes the colour”. (1) The lesion often fills a small part of the frame, so whole-image statistics mostly measure the surrounding skin: skin tone, sun damage and the redness of the body site, while site and age differ by class (actinic keratoses and SCC on sun-damaged heads of older patients). (2) Acquisition: dermoscope (contact plate, polarised light, dark corners, rulers, ink) vs. camera, plus lighting, white balance and devices that differ by clinic while the class mix also differs by clinic (see the `image_manipulation` and missing-site patterns). (3) Sampling: the sample is balanced per class, but the boxes of DF, INF, VASC, BEN_OTH (30 lesions) and MAL_OTH (9) are noisy.

Conclusion. Colour and brightness alone are a weak signal: the distributions overlap heavily, the best single feature reaches AUC 0.66, and part of that comes from class composition and acquisition rather than pathology. The project needs features that capture spatial structure (borders, pigment network, vessels, texture), i.e. a CNN, with only mild colour augmentation so the little colour signal is not erased. Because the image-type gap is larger than any class gap, both image types must be represented in every class and split, or handled explicitly.

3. Reusable pipeline (Parts 3–5)

Design. All code is one importable package, `milk10k/`, with a module per part: `preprocessing`, `loader`, `visualize` (and `metadata_analysis`, `color_analysis` for Parts 1–2). `config.py` is the only place with paths (environment variable `MILK10K_DIR`, default `data/` in the repository, so no absolute paths), the seed (42) and the image size (224). Scripts and notebooks only *call* these functions, so later milestones extend them (splits, torchvision transforms) instead of copying code. Eleven pytest tests check shapes, value ranges, skipping of bad files, the available-only filter and reproducible shuffling.

Preprocessing. `preprocess_image(source, size=224, color_mode="rgb"|"gray"|"hsv", normalize="scale"|"minmax"|"zscore"|"none", keep_aspect=True)` takes a file path, uint8 array or PIL image and returns (array, info): a float32 (H, W, C) array plus a dict with the original size, final shape, options and value range. Resizing scales the shorter side to 224 and cuts the centre square, so lesion shape (asymmetry, border) is not distorted; the trimmed edges are mostly skin and dark dermoscope corners. Default normalisation divides by 255, i.e. min-max scaling with the fixed 0–255 range: it fits no statistics (no test information can leak), keeps real brightness differences between images, and matches torchvision’s `ToTensor`. Per-image min-max would stretch every image’s contrast and erase those differences (Figure 5, last column); z-score is available with a mean/std that must come from the training split. `preprocess_batch(items)` takes metadata rows (DataFrame) or a list of paths and returns `BatchResult(images (N, H, W, C), ids, infos, skipped)`. A missing or corrupt file is skipped with its reason (demo: 3 processed; 2 skipped with `FileNotFoundError` and `UnidentifiedImageError`); a wrong option raises at once instead of silently skipping every image.

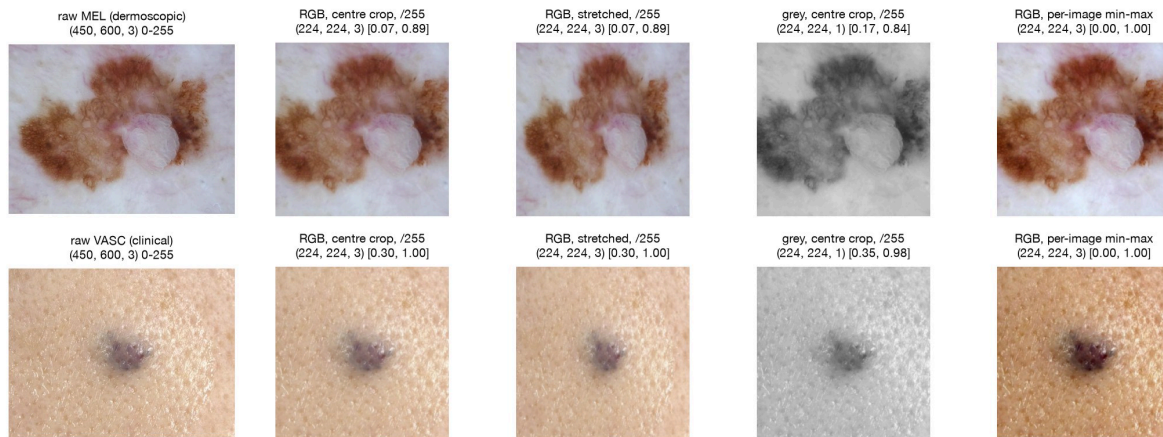


Figure 5. Raw image (left) and processed versions with shape and value range: centre crop vs. stretch, greyscale, and per-image min-max, which changes each image's overall colour.

MilkLoader	Interface (Part 4)
Construct	<code>MilkLoader(metadata, img_dir=None, label_col="diagnosis_1", batch_size=32, shuffle=True, seed=42, classes=None, drop_last=False, **preprocess_options)</code>
Available only	keeps the rows whose .jpg exists on disk (10,480 of 10,480 here); <code>n_unavailable</code> counts the rest
Iterate	<code>for images, labels, ids in loader:</code> → images (B, 224, 224, 3) float32 · labels (B,) int64 · ids = list of <code>isic_id</code> ; one pass = one epoch
Labels	<code>loader.classes = ["Benign", "Indeterminate", "Malignant"]</code> → 0, 1, 2
Memory, speed	images are read and processed batch by batch, never all at once; 1.9 ms per image, about 20 s per epoch
Order	new shuffle every epoch from <code>seed + epoch</code> : different each epoch, identical across runs
Helpers	<code>len(loader)</code> = batches per epoch (328 at batch size 32); <code>class_counts()</code> ; <code>skipped = (id, reason)</code> pairs

Visualizer. `show_grid` plots images with their labels as titles from a list of arrays or straight from a loader batch (Figure 7); `plot_class_balance` draws counts and percentages per class (Figure 6); `plot_batch_summary` plots the per-channel value distribution of a processed batch next to raw/processed pairs, to confirm the normalisation (values stay within [0, 1]; saved as `figures/session2/fig8_batch_summary.jpg`).

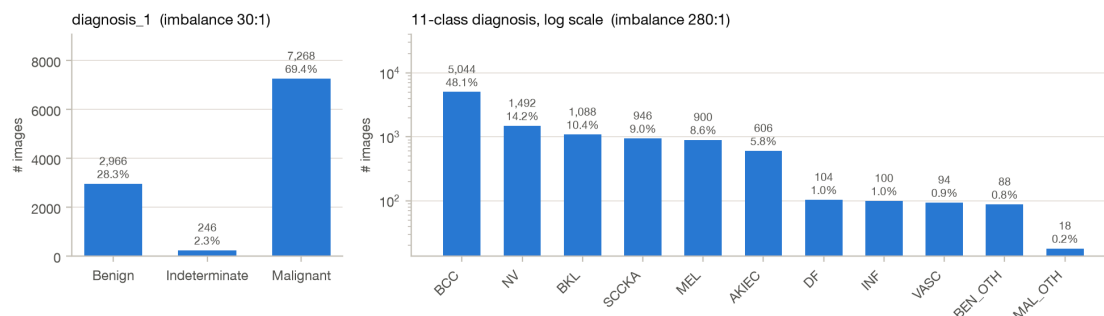


Figure 6. Images per class (each lesion contributes 2 images).

Class balance. The data are strongly imbalanced: 69.4% of images are Malignant and 2.3% Indeterminate (30 : 1); among the 11 classes BCC alone is 48.1% and MAL_OTH 0.2% (18 images, 280 : 1). All 123 Indeterminate lesions are actinic keratoses (AKIEC), a class that also holds 180 Malignant lesions. Consequences: accuracy misleads (always answering “Malignant” scores 69%), so I will report balanced accuracy, macro-F1 and per-class recall; splits must be stratified and grouped by lesion; training needs class weights or oversampling; the rarest classes may need merging. The imbalance reflects biopsy-based collection, not real-world prevalence.



Figure 7. First batch from `MilkLoader` (batch size 16), drawn with `show_grid`.